

Dart Emerald

The Dart Emerald sports a 2 GHz Pentium 4 processor combined with an Intel 845GL-based motherboard. It comes with a 40 GB Samsung IDE hard disk and 256 MB of DDR RAM, a part of which is used by the onboard Intel Extreme graphics. There is no AGP port provided on the motherboard and therefore, there is no room for enhancing the system's graphics performance.

The onboard SigmaTEL integrated chipset sings to you through the 2.1 speakers and the AT keyboard comes with a PS/2 converter and features very good tactile response, sans any shortcut keys.

One area where the Dart Emerald makes a mark is in the package contents. It provides you with a number of educational, game and multimedia titles. Also provided are all the necessary driver CDs, along with Windows XP Home Edition. The Dart Emerald is sold with a three-year onsite warranty. At Rs 40,000, it's quite suitable for home users.

Suitability to Task Index	
Gaming	5.58
Productivity	6.93
Office apps/Internet	5.06

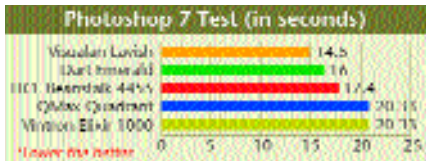
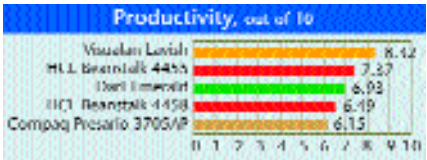
with its 2 GHz Pentium 4 processor, but featured a standard 40 GB 5,400-rpm Seagate hard disk drive coupled with 128 MB of DDR RAM and the integrated graphics of the 845G chipset. It scored reasonably well in the SiSoft Sandra tests and the inclusion of the 2.1 Altec Lansing speakers with this system enabled it to do well in the audio tests.

Understandably, the computers that fell below par in this subtest were the ones based on the Celeron/Intel 810 combination—the Vintron Elixir 1000 and the Visualan Celebrity. Since they featured slower processors and integrated graphics systems that used shared memory, their performance in the gaming and the low-level subsystem tests were among the lowest in this category.

Productivity: For this application area, we looked at the computer's performance in running tasks such as office applications, image editing, video encoding, etc. Therefore, we observed the performance of the computer's processor, hard disk and memory along with real world applications such as Photoshop, Virtual Dub video encoding and the performance of subsystems such as the monitor, CD-ROM drive and soundcard.

Once again, the computer that came out tops was the Visualan Lavish. The components that were responsible for its success were the 2.53 GHz Pentium 4 processor, 256 MB of DDR RAM and the fast hard disk drive. This was clearly seen in the Photoshop test where it crunched through the set of 21 filters in just 14.5 seconds. Also, since Photoshop 7.0 had special support for the Pentium 4's SSE2 instruction set, the performance of the computer in this application was unsurpassed. Even in the video encoding tests, the Visualan Lavish delivered stellar scores by processing the test video file in just 1 minute 3 seconds. The 17-inch monitor was clear and sharp and quite suited to multimedia applications that require attention to detail. The accompanying CD-RW drive burned a complete 650 MB CD in a span of just 3 minutes 18 seconds. All this makes the Visualan Lavish suited to authoring multimedia titles for distribution, image editing and video processing.

Next up, the HCL Beanstalk 4455 did very well in the processor, memory and especially the hard disk subsystem tests. The memory bandwidth of its 256 MB of DDR RAM was logged at a shade under 2 GBps, making it well-suited to applications such as image and video editing where large files are stored and worked upon in RAM. With its RAID-0 implementation of the two 40 GB Seagate Barracuda hard disks, the system delivers brilliant data transfer performance, especially in the sequential read and write tests (43 MBps and 40 MBps, respectively). This implied that it is a good choice for applications such as audio and video



...to Keep the Doctor Away

Since computers comprise an assortment of delicate components, regular cleaning and maintenance is necessary to ensure your system's long life. The following are the main areas you should concentrate on for keeping your PC in optimum shape:

- You hard disk is most susceptible to wear and tear and one of the best ways to maintain it is to nip any problems in the bud. Regular system scans will ascertain that the file system is in good condition and will take care of the minor problems that may arise as a result of crashed programs and hard reboots. To ensure that the space on your hard disk is well utilised, you should defragment it regularly. This arranges the files in contiguous clusters, thus improving performance and freeing up wasted space.
- Hardware manufacturers usually have product Web sites that

offer regular driver updates that improve performance or iron out hardware compatibility issues and bugs. Update your drivers frequently for a smooth-running system. The same logic extends to patches for software applications such as your operating system, office suites and anti-virus packages. As a precaution, verify that you load the updates and drivers specific to your operating system.

■ Don't forget to keep your system clean and free from dust. While not perceived as an immediate threat, dirt and dust are silent agents that cause serious problems down the line. They affect the moving components in your system: from the keyboard keys, to the more critical components such as the fans cooling your system. Covering and cleaning your computer will help ensure that these critical system components do not fail due to a buildup of dust.